

Analog Lab 3

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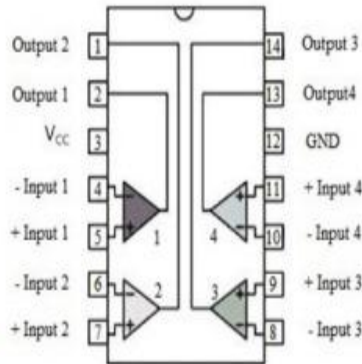
Objective :-

The objective of this experiment is to build a half bridge driver and check a d class amplifier with the help of PNP and NPN transistors. We will optimize the base drivers using inverter buffers. We are going to simulate the circuit using LTSpice then going to build a physical circuit to check our outputs.

Components Required :-

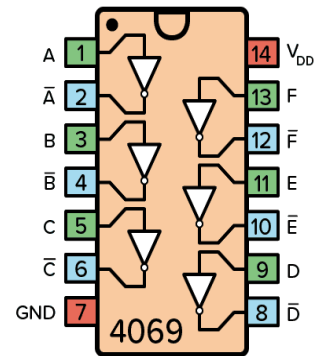
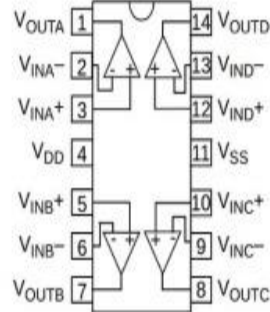
1. Pulse width modulator circuit (week 2)
2. 74LSHC00N for NAND Gates
3. CD4096 for Inverters.
4. PNP (2N2907) and NPN (2N2222)
5. Capacitors and resistors.

LM339



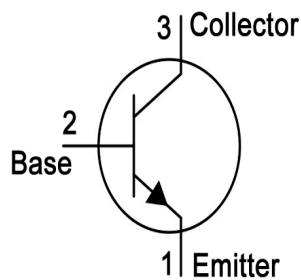
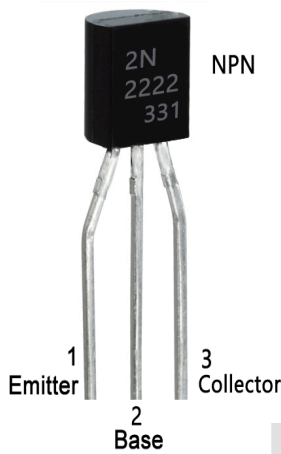
MCP6004

PDIP, SOIC, TSSOP



2N2222 Transistor Pinout

TO-92 Package

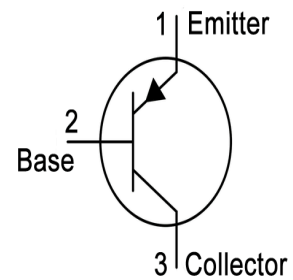
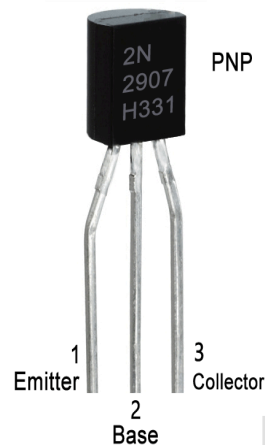


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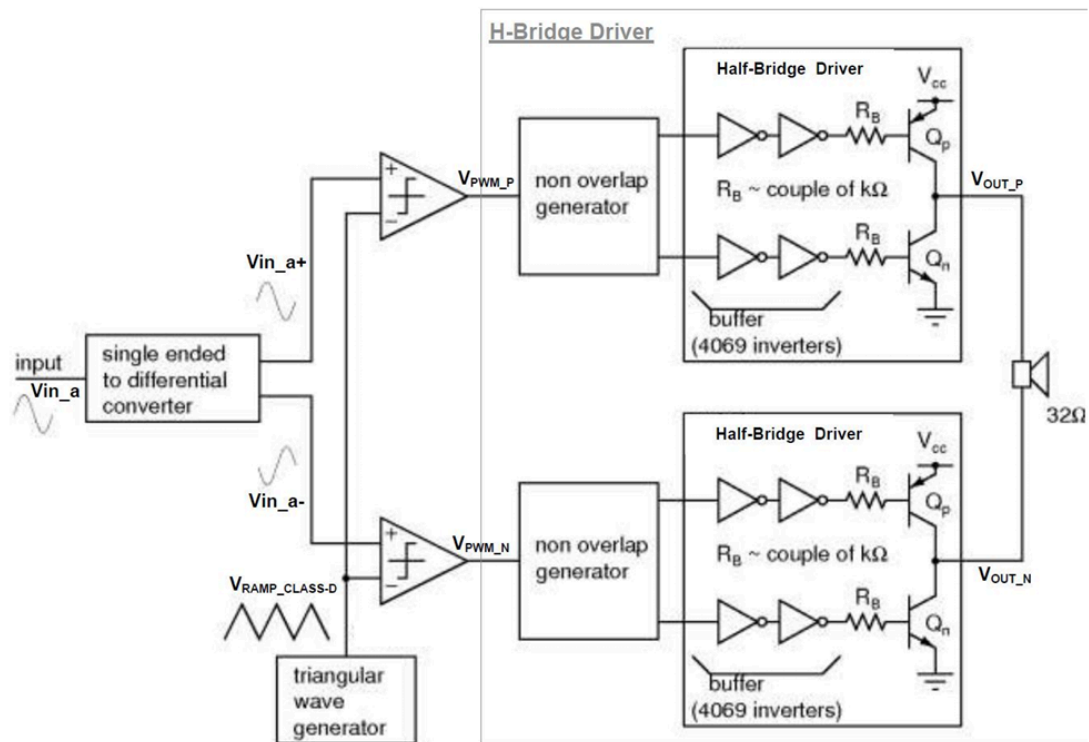
2N2907 Transistor Pinout

TO-92 Package



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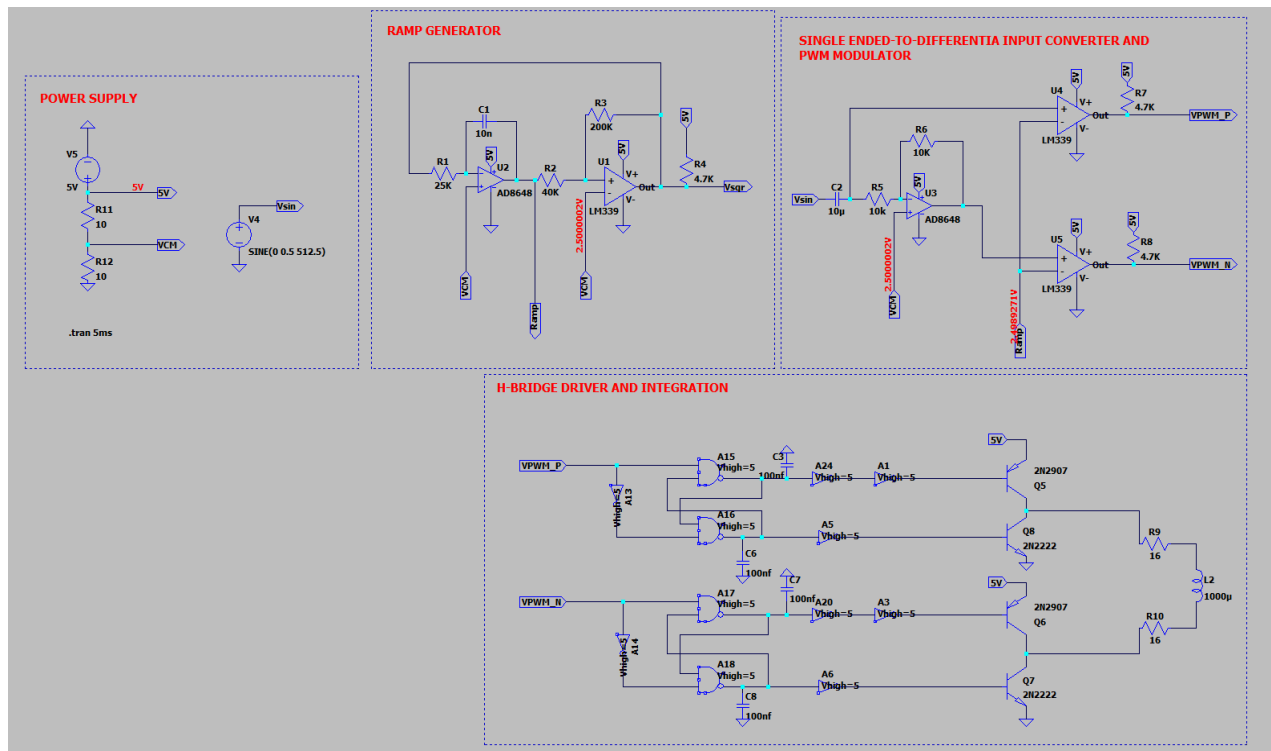
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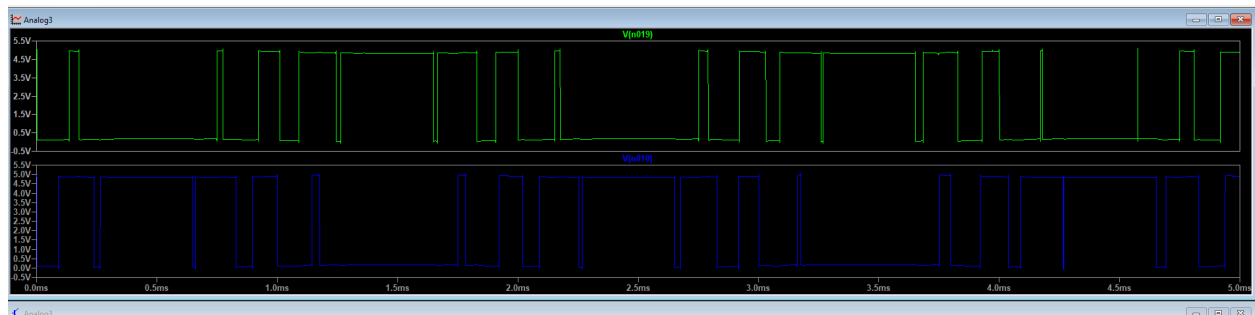
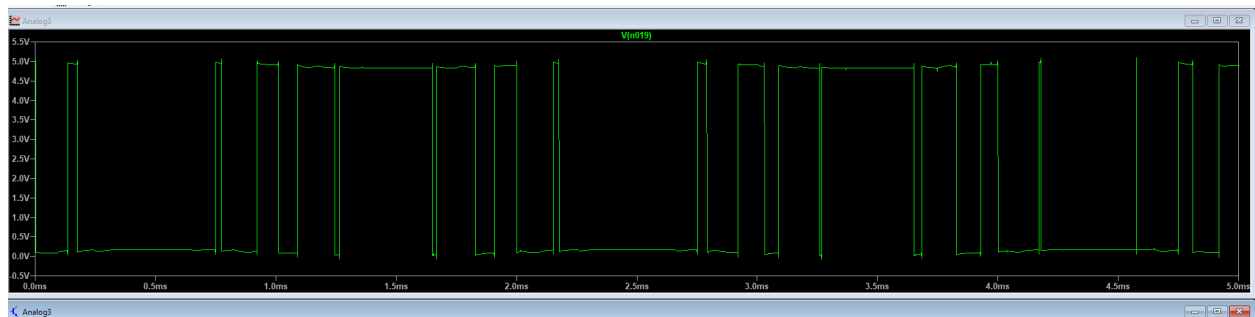
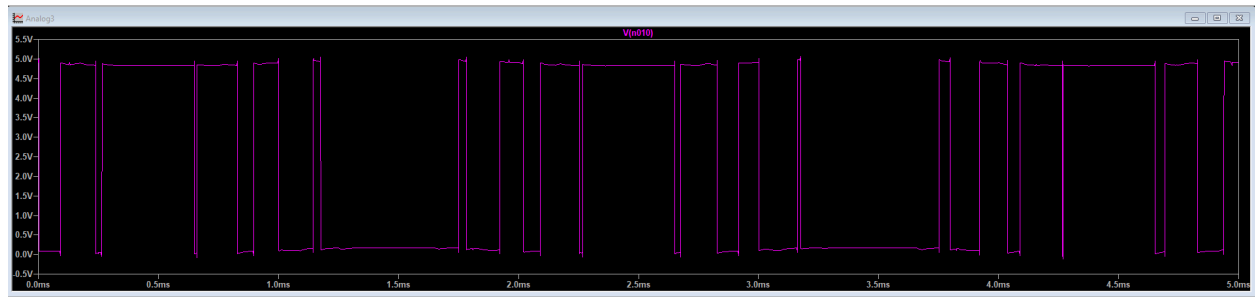
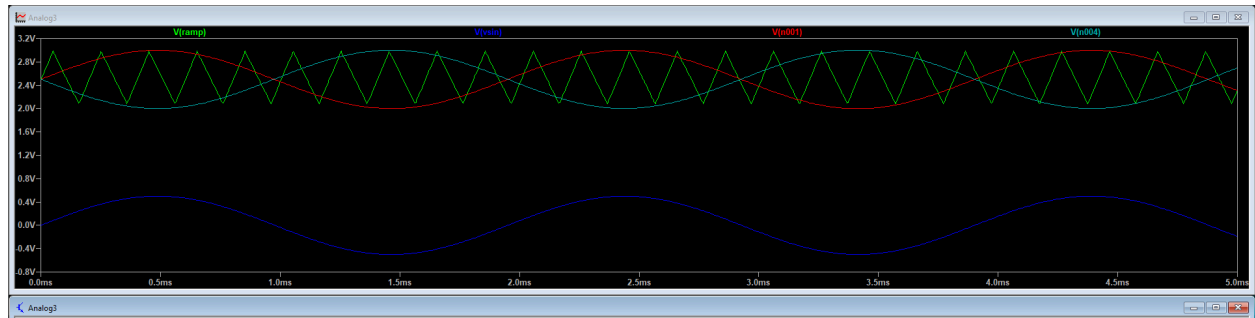
Note: triangular wave and signal inputs to the comparator must be around the same bias point

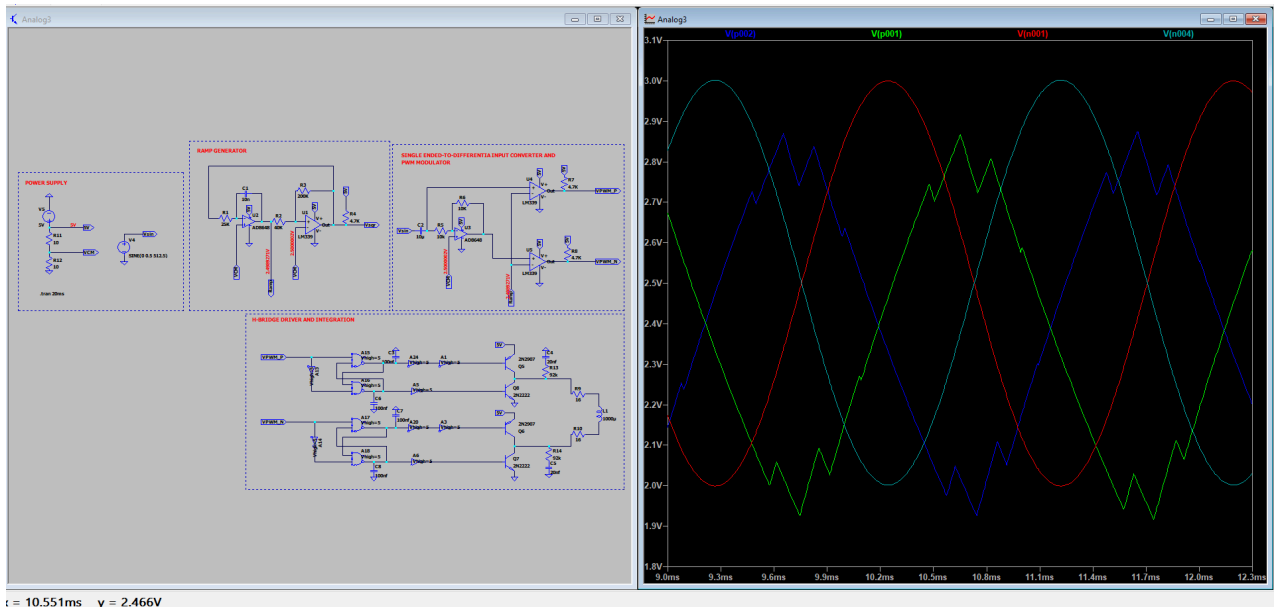
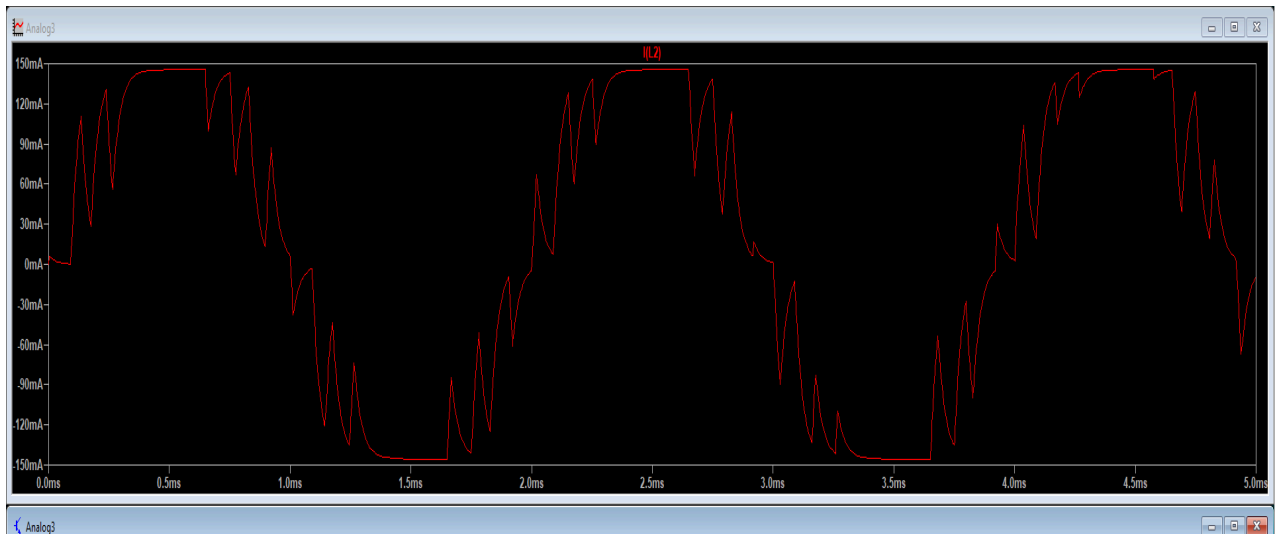
Figure 7 Circuit diagram of a complete class-D amplifier.

LT Spice Circuit :-

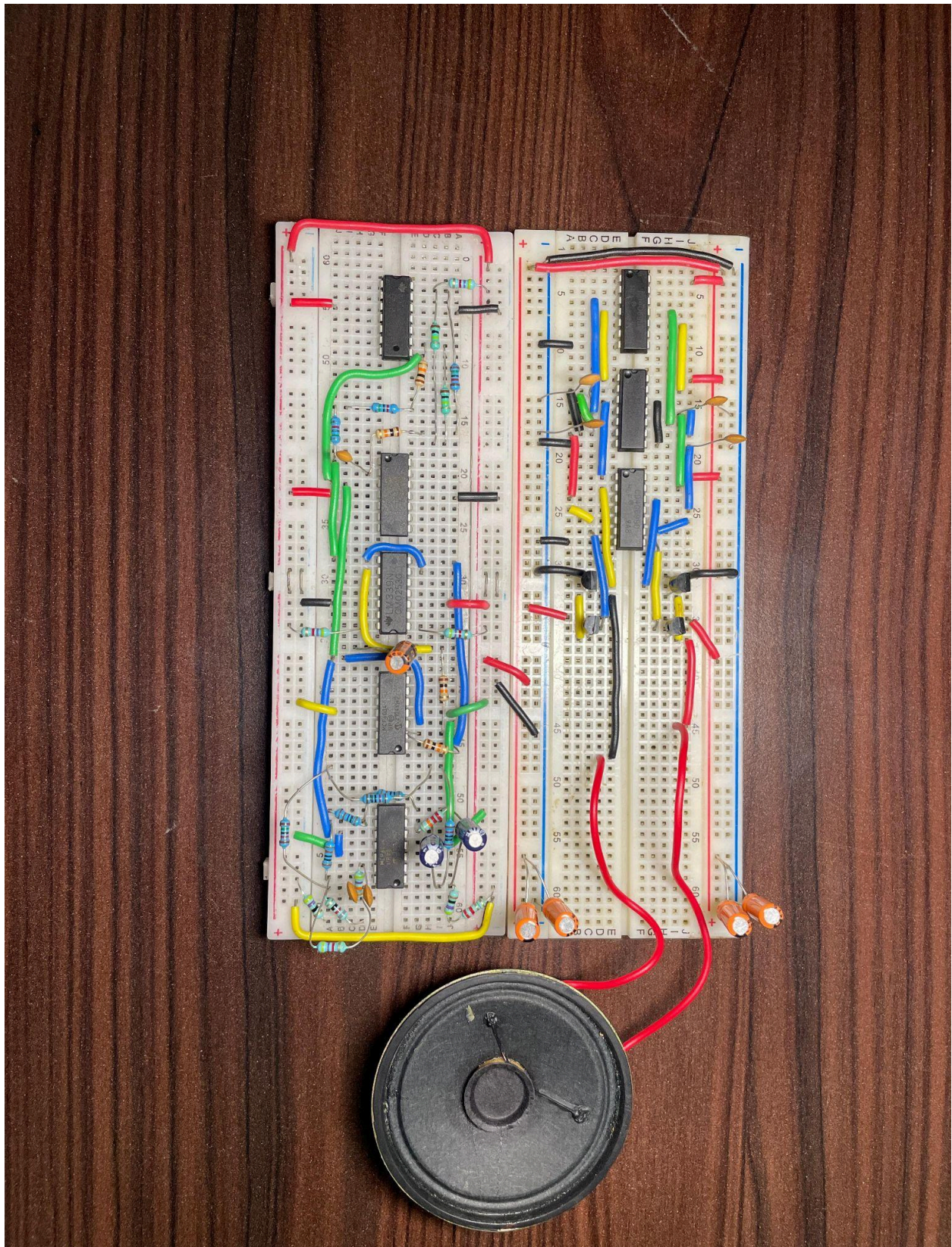


LT Spice Output :-

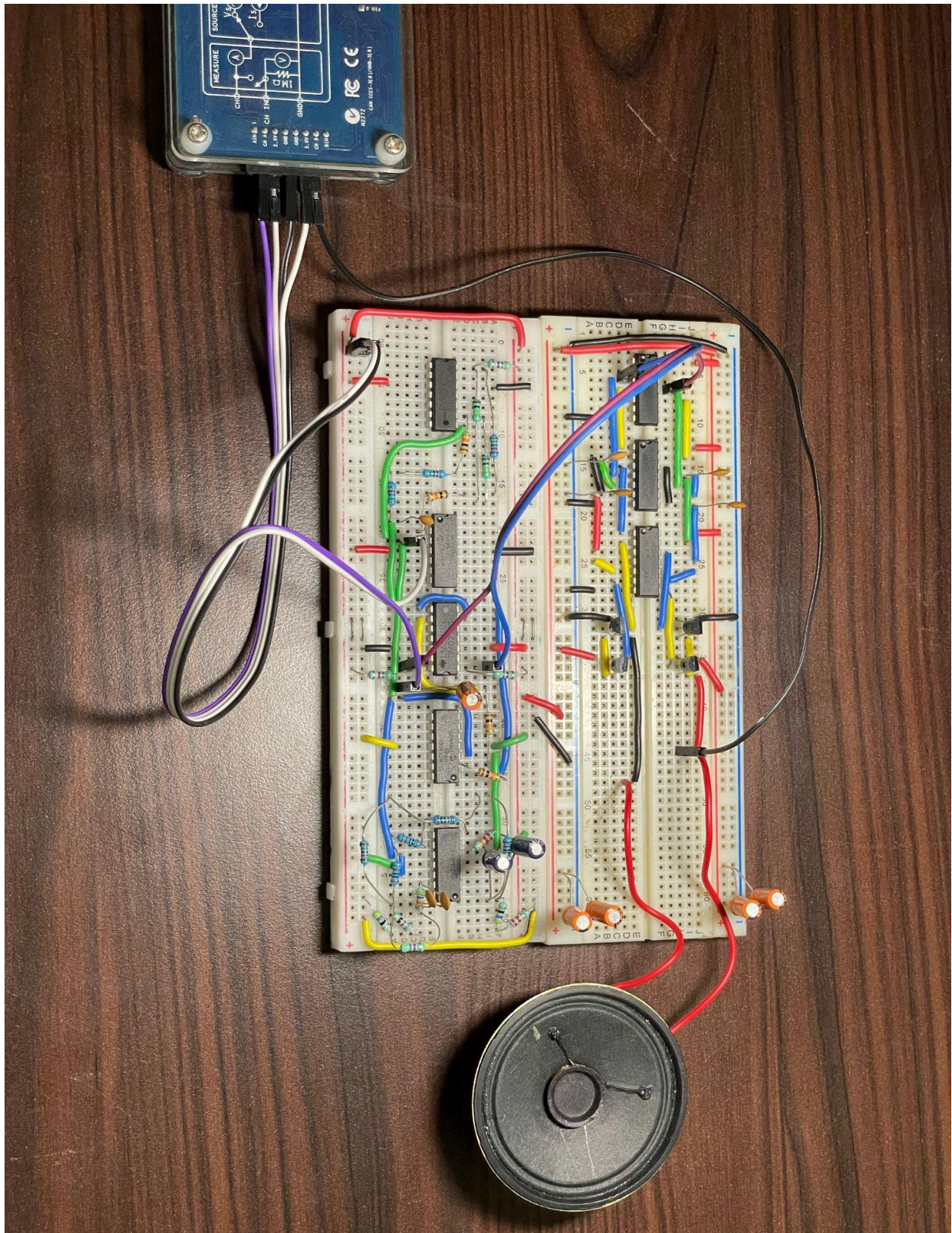




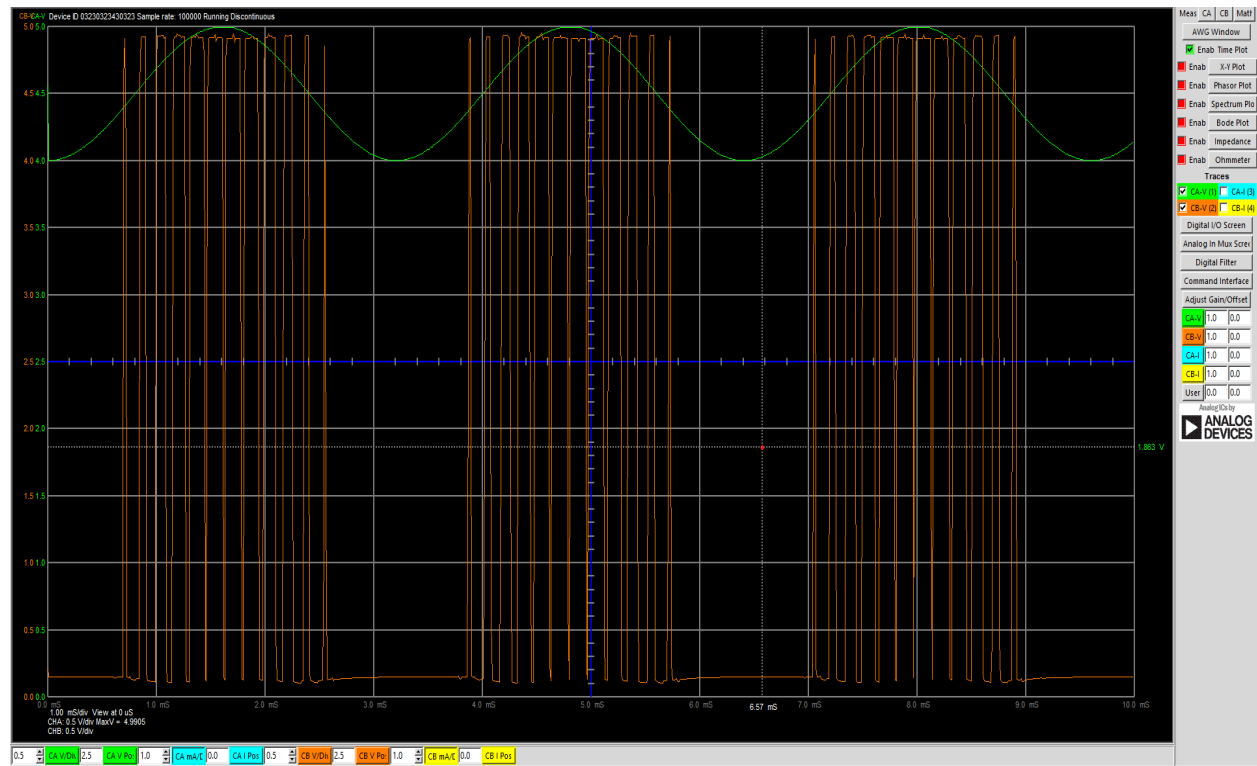
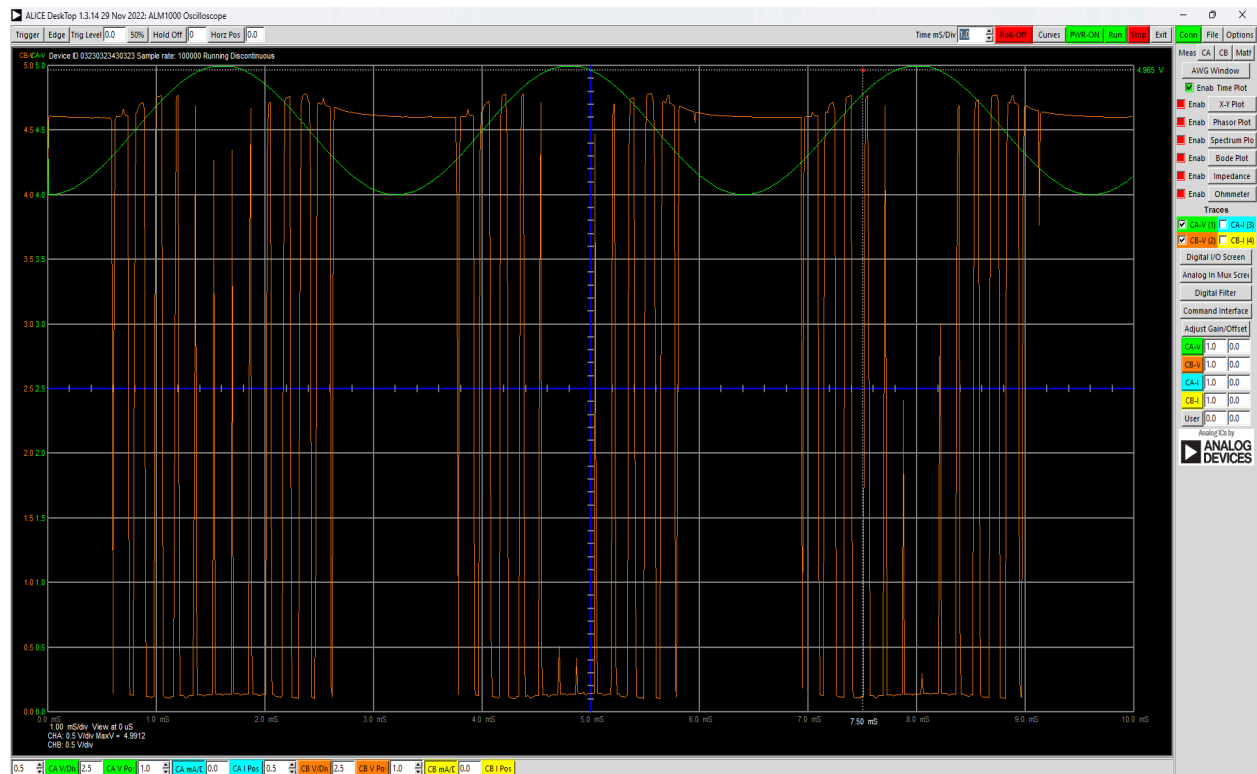
Physical Circuit On Breadboard



Physical Circuit with ADALM 1000



Input VPWM_P And VPWM_N Signal



Output VOUT_P And VOUT_N Signal.

